Chapter 17

Applications Of Inversive Geometry

Verbal College Entrance Exams

Back in 1980-ies Moscow, then USSR, the entrance exams into the top technical schools of higher eduction usually consisted of three distinct parts. Part 1 was an examination on mathematics, Part 2 was an examination on physics and Part 3 was an examination on literary arts which amounted to writing a composition based on a given prompt.

Further, a standard college entrance examination on mathematics and physics consisted of two parts, a written exam followed by a verbal exam. If a perspective student failed a written examination then for that person the game was over for that particular school. If a perspective student managed to pass the written exam then (s)he would be allowed to take the verbal exam.

If that person failed the verbal exam then, again, for that perspective student the game was over for that school and (s)he would have to look for a different potential college. A success on a verbal exam meant that the perspective student would be allowed to move on through the examination filter until (s)he either failed the next examination or successfully passed through all the remaining exams.

In such an elimination-on-first-failure process the examination on mathematics was always carried out first, for mathematics is a foundation of all other technical disciplines. If a potential student failed any one of the mathematical exams then there was no reason to proceed any further.

As such, the importance of knowing as much elementary mathematics as possible for the hopeful students of technical schools could not be overestimated and practicing solving problems in mathematics was paramount.

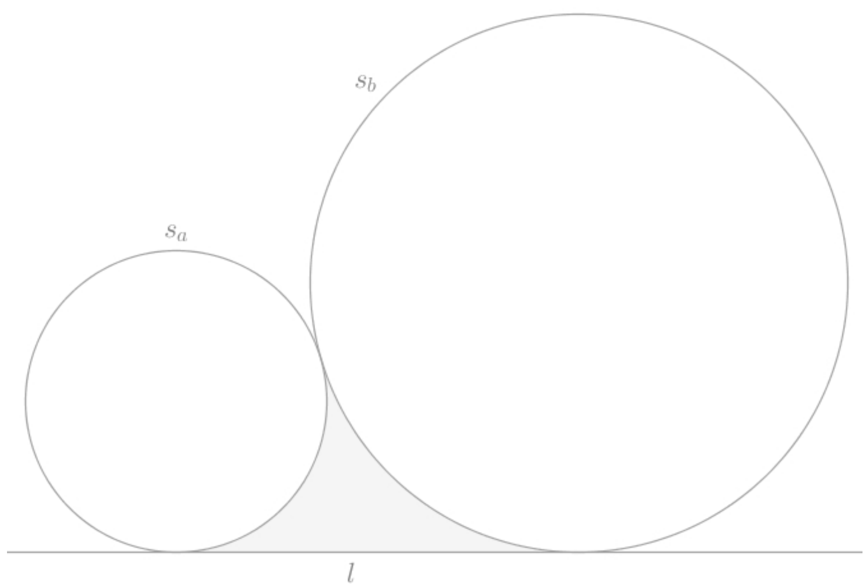
The overall goal of these written and verbal mathematics and physics exams, during which the future students were expected to show the development of their solution blow-by-blow and step-by-step, was for the members of the admission committee to see how a person thinks and sometimes, on a rare occasion, of course, it was possible to get a passing score even without actually obtaining a correct answer if the line of reasoning presented was original enough.

In particular, for a written examination in mathematics (and physics) the future students were expected to turn in all their sheets of paper with the detailed explanation of how they obtained the answers for all the problems that they managed to solve.

During a verbal examination in mathematics (and physics) the examiner had the right to deep-dive in real time into any topic that was relevant to the given problem.

What follows below is a representative sample of a problem that could have, and very likely at some point was, given during a verbal mathematical college entrance examination.

Problem 24. Two coplanar circles, $s_{a,b}$, with the radii $r_a < r_b$ touch each other externally (Figure 17.1):



while touching a coplanar straight line l.

Objectives: a) inscribe a circle, s_c, into the gray curvilinear triangle trapped between the circles $s_{a,b}$ and the straight line l (using the Euclidean tools alone, of course)

b) find a relationship that ties the lengths of the radii $r_{a,b,c}$ of all three circles $s_{a,b,c}$ together

Solution. In addition to showcasing the constructive potential of inversive geometry, we will use this simple problem in order to discuss, in some depth, the mechanics of the reverse-order problem-solving approach.

A newcomer to the art of problem-solving, who is witnessing how *other people* solve seemingly difficult problems, may ask a question:

- It looks like these *other people* follow some trail that is completely invisible to me, but, sure as hell, they do get to their destination! What is this invisible trail? And how come I cannot see it?

So far we have been using this invisible trail all over the place but we never explicitly stated so and we never explained the exact mechanics of the trail that we refer to as the reverse-order or reverse-timeline problem solving approach, until now.

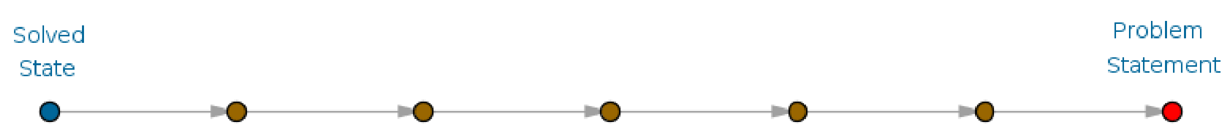
Here is one way to explain how the said approach works and can be used.

When we want to test whether a given problem will fall under the reverse-order or reverse-timeline attack, we first assume that the problem at hand has been solved.

Exactly how that problem has been solved is of no concern to us at that stage of the process. For all that we care, an all-knowing Mathematical Oracle had magically arranged the pieces of the puzzle in such a way that the said puzzle is solved completely.

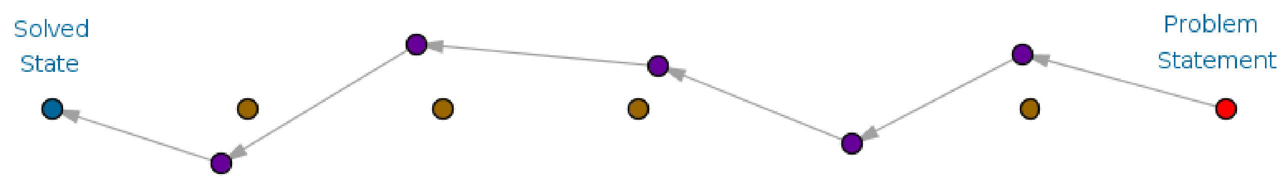
We, thus, begin the problem-solving process with the given problem being in the solved state. After the problem-has-been-solved assumption is made, we try to synthesize a string of correct IF P THEN Q implications that originates at the said solved-state and then, chewing on the intermediate implications, moves toward and, finally, terminates at the original problem statement or the problem's unsolved state.

If we were to model such implications with geometric points and the flow of causality with vectors then a typical successful journey described above might look as follows (Figure 17.2):



There are no guarantees of any kind here for everything that we are discussing at the moment has a big IF in front of it.

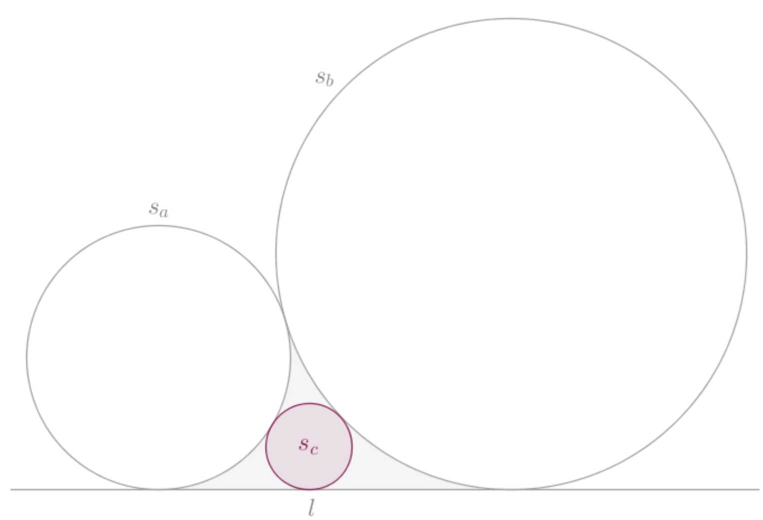
Thus, if, and that is a big if, so if, starting at the problem's solved state, through a string of correct implications, we manage to arrive at the original problem statement then, as the lightweight theory goes, it is plausible that on the said reverse-order journey we may collect enough evidence or enough tactical material, insight and wisdom that will allow us to construct a successful proof or a computation, as the case may be, as we move in the opposite, forward-order or the direct and expected flow of causality, direction from the problem statement to the solved state (Figure 17.3):



Above, we have changed the color and the arrangement of the forward-order implications because these implications are not the same implications that we used in our reverse-order reasoning. We have left the reverse-order implications in the diagram above to show that they now play the role of guiding land-marks or lighthouses.

Separately, observe that our description of the reverse-order problem-solving approach lacks any technical jargon and it can be taken not only to any, vaguely defined, branch of mathematics but to (theoretical) physics, computer science and computer programming and beyond.

Thus, working in reverse order, we assume that the given problem **a)** has been solved (Figure 17.4):



Exactly how this problem has been solved we do not know yet and we do not care at the moment. We simply *assume* that the circle s_c has been inscribed into the gray curvilinear triangle as demanded.

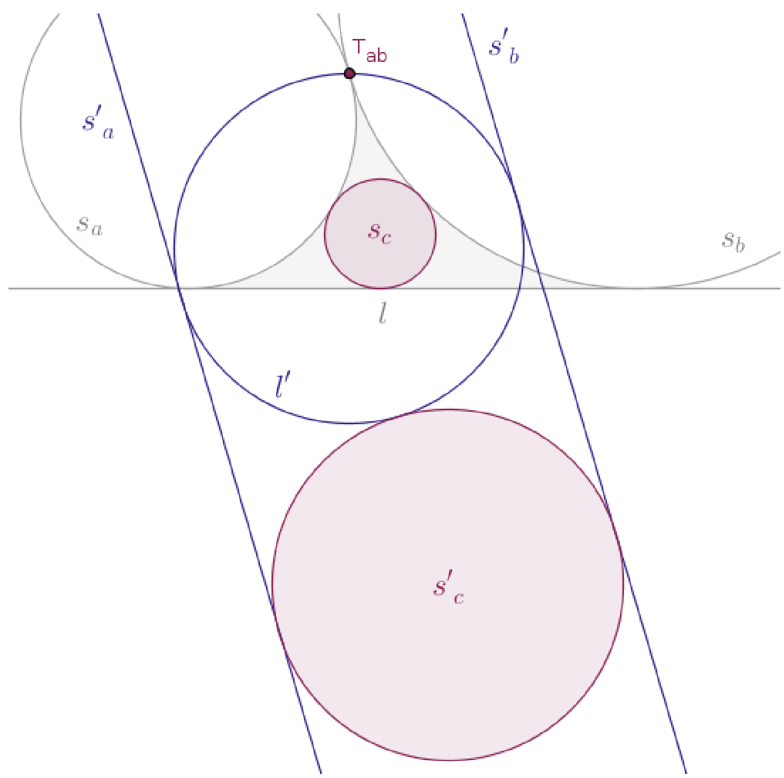
Taking that solved state as the starting point, what string of correct IF P THEN Q implications will bring us to the original problem statement? Formally, when we turn the above solved state into a premise, an antecedent or a hypothesis:

if the four given objects touch as shown

then what could a winning conclusion or consequent be?

Well, this is where the transformation that we are currently studying comes in handy. If the four given objects touch as shown then under an inversion of the parent plane with respect to a circle the inverse images of the four given objects will also touch. That much we already know and that much we already proved.

Which specific inversion though? An inversion that has its center at the point, T_{ab}, where the circles $s_{a,b}$ touch. Why is that? Because earlier we proved that any circle that passes through the center of inversion is taken into a straight line that does not pass through that center and conversely. Any straight line that does not pass through the center of inversion is taken into a circle that does pass through that center (Figure 17.5):



Thus, in this case, under the inversion of the parent plane, with positive power, with respect to a circle $c(T_{ab}, r)$, not shown below in order to avoid the clutter:

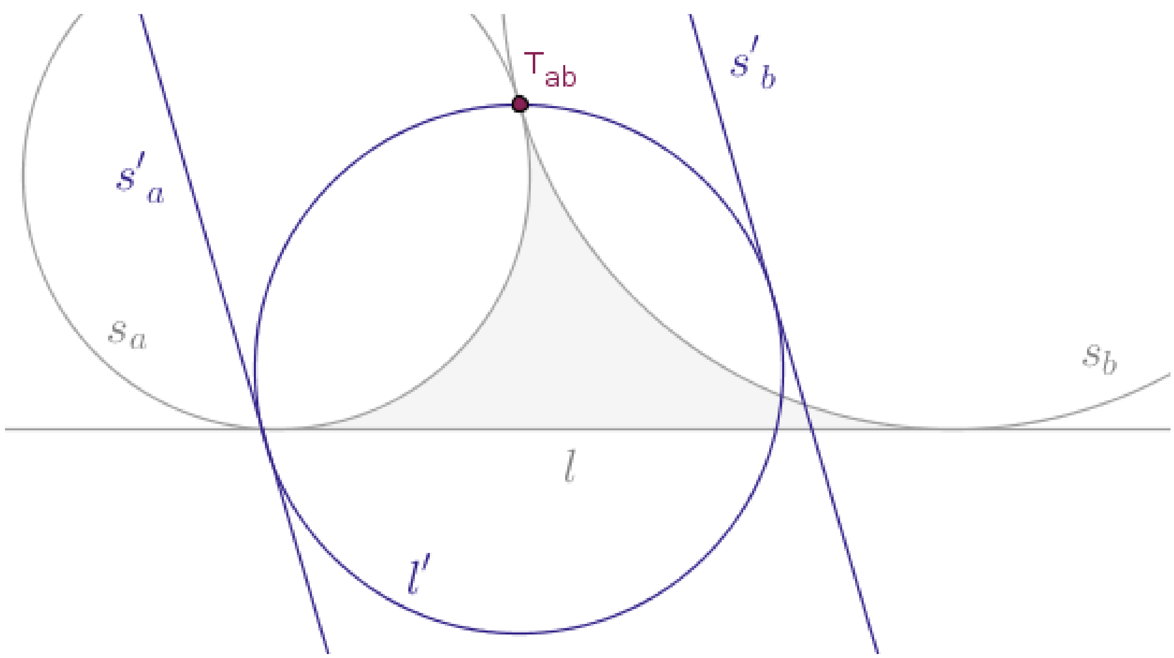
- the externally tangent circles $s_{a,b}$ will be taken into two corresponding straight lines $s'_{a,b}$ that will be parallel to each other
- the straight line l will be taken into a circle l' that will pass through the center of inversion and will touch both parallel straight lines $s'_{a,b}$
- the mystery circle s_c will be taken into a circle s'_c that will still touch:
    - the parallel straight lines $s'_{a,b}$ and
    - the circle l' externally

and we are done. Why. Because if we now remove the circles s_c and s'_c from the picture then we will arrive at the original problem statement **a)** and, to boot, its inversive equivalent will be stated as follows:

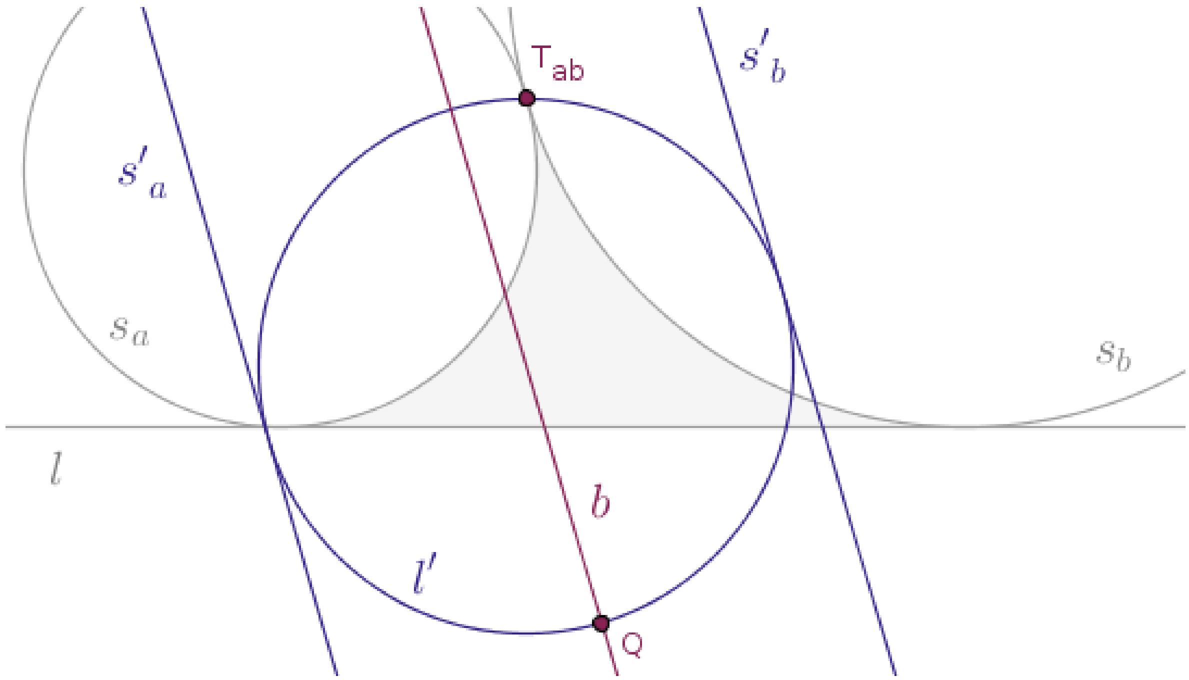
construct a circle (s'_c) that touches two given parallel straight lines (s'_{ab}) and a given circle (l') externally, where the given circle itself (l') touches the said parallel straight lines (s'_{ab})

But the above task is primitive in the sense that it can be carried out with Euclidean tools alone and it can be qualified as a previously-solved problem. Game over. Officially, as our approach prescribes, we now flip the direction in which the relevant causality flows to its opposite and we tell the examiner our solution of the problem **a)**:

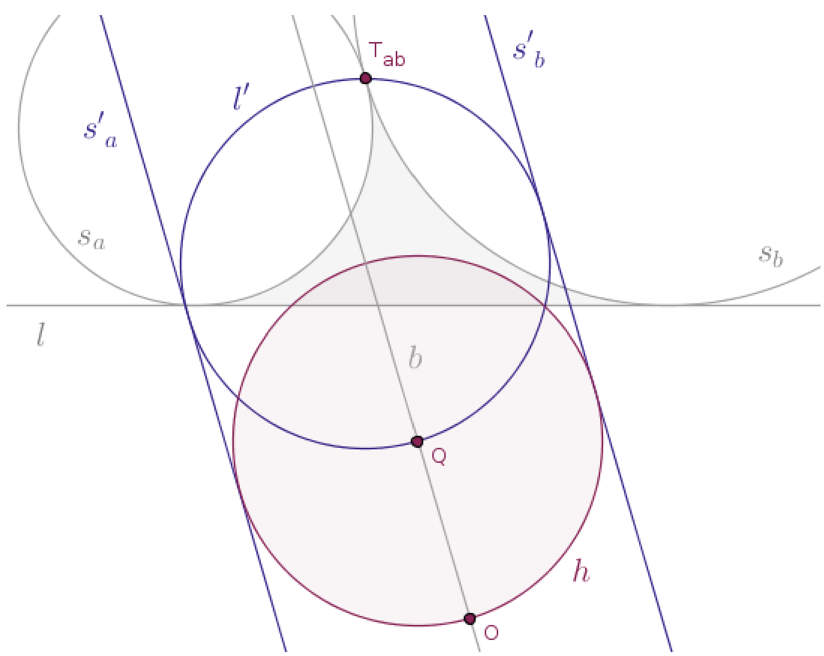
Step 1. Reflect the three given objects in a circle $c(T_{ab}, r)$ with positive power to obtain their corresponding inverse images (Figure 17.6):



Step 2. Draw a straight line, b, whose every point is equidistant from the parallel straight lines s'_{ab} until b intersects the circle l' at a point, Q, farthest from the center of inversion (Figure 17.7):

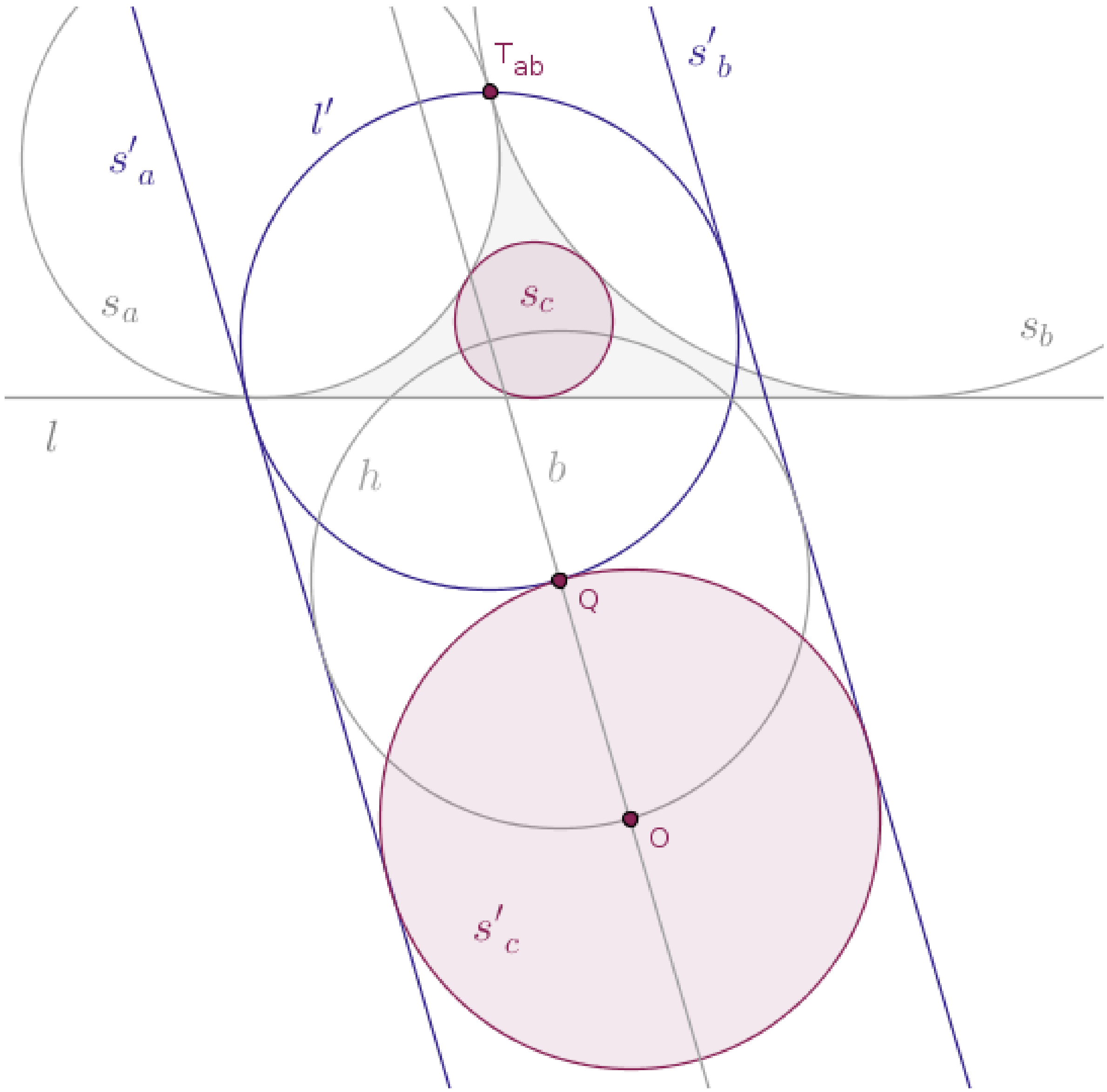


Step 3. Draw a helper circle, h, centered at the point Q with the radius whose length is equal to the length of the radius of the circle l', which is also exactly half the distance between the parallel straight line $s'_{a,b}$, which was obtained in the previous step (Figure 17.8):



The helper circle h will intersect the straight line b at two distinct points, O and P. Let O be the point that is farther away from the center of inversion than the point P.

Step 4. Draw a circle, s'_c, centered at the point O with the radius $|OQ|$ and invert it in the circle $c(T_{ab}, r)$ with positive power into the sought-after circle, s_c (Figure 17.9):



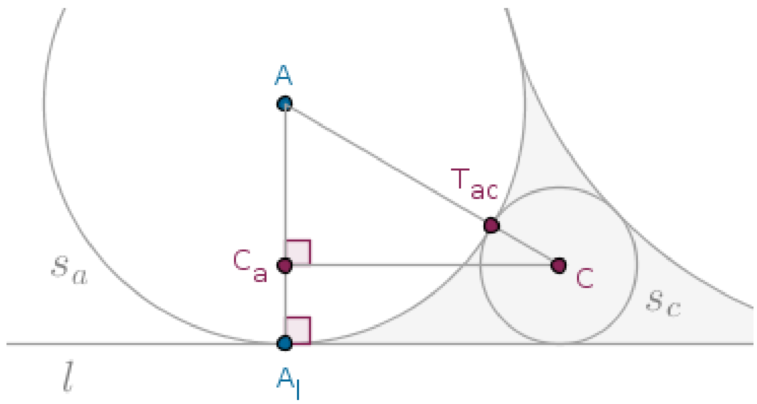
The inversion of the parent plane described above solves the problem **a)** completely and correctly: the circle s_c thus produced will touch the given circles $s_{a,b}$ externally while touching the straight line l.

As it is the case with some constructive problems, the very construction of the solution object is, pretty much, a proof that the object thus produced is, indeed, the one sought-after.

Observe that the inversion of the parent plane that we described above is not the only that will solve the given problem. There exist other inversions of the same plane that will solve this problem as well and we leave it as an exercise to our readers to figure out which some of these inversions are on their own.

Now that we know how the requested circle s_c comes into existence, we can address the second part of the problem, **b)**, and here it quickly becomes evident that in order to unmask a relationship across the radii $r_{a,b,c}$ of the circles $s_{a,b,c}$ all we have to do is use the Pythagorean theorem a few times.

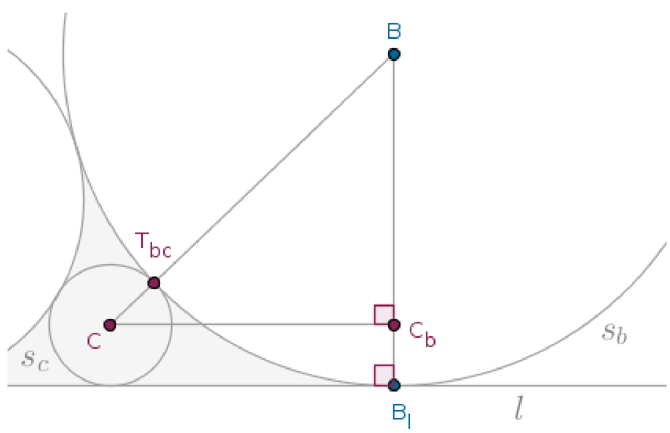
That is. Let the point A_l be the orthogonal projection of the center, A, of the smaller circle, s_a, onto the given straight line, l, and let C_a be the orthogonal projection of the center, C, of the inscribed circle, s_c, on the line segment AA_l. Then, from the right, by construction, triangle AC_aC (Figure 17.10):



we, clearly, have:

$$|C_aC| = \sqrt{(r_a + r_c)^2 - (r_a - r_c)^2} = 2\sqrt{r_a \cdot r_c}$$

and rom the right, by construction, triangle BC_bC (Figure 17.11):

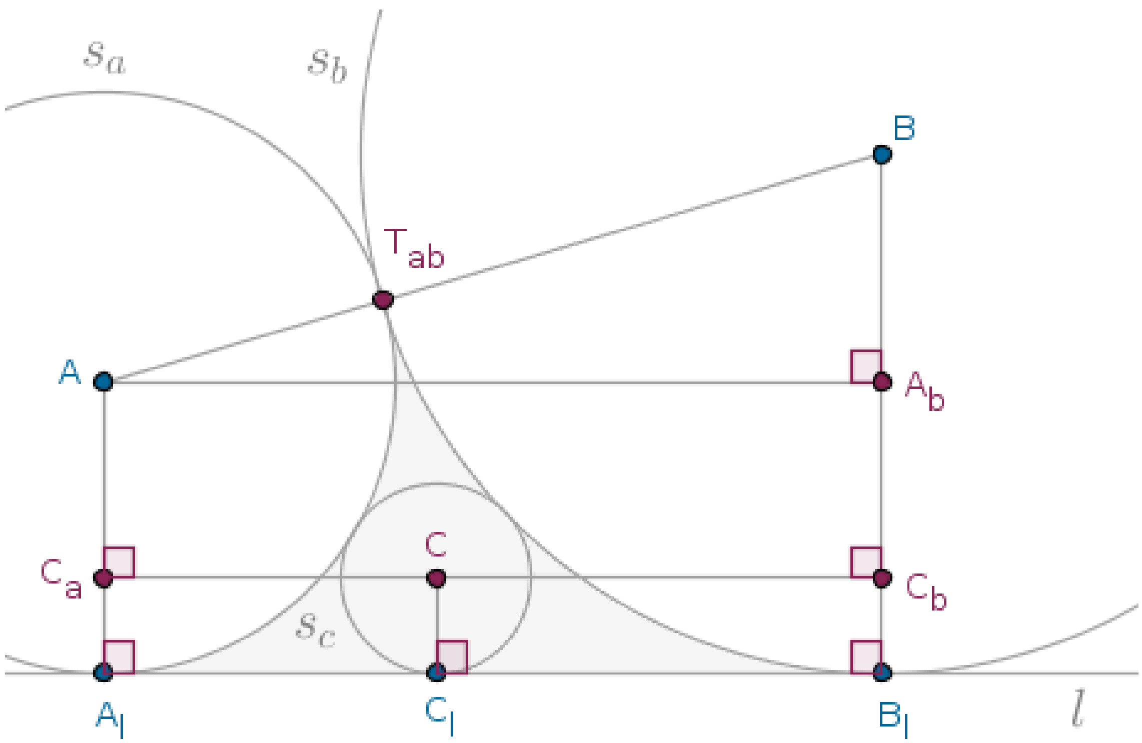


we, clearly, have:

$$|C_bC| = \sqrt{(r_b + r_c)^2 - (r_b - r_c)^2} = 2\sqrt{r_b \cdot r_c}$$

if we agree that B_l is the orthogonal projection of the center, B, of the larger circle, s_b, onto the given straight line, l, and that C_b is the orthogonal projection of the center, C, of the inscribed circle, s_c, onto the line segment BB_l.

And if A_b is the orthogonal projection of the point A onto the line segment BB_l then from the right, by construction, triangle AA_bB (Figure 17.12):



we, clearly, have:

$$|AA_b| = \sqrt{(r_b + r_a)^2 - (r_b - r_a)^2} = 2\sqrt{r_b \cdot r_a}$$

But since it is the case that:

$$|AA_b| = |A_lB_l| = |A_lC_l| + |C_lB_l|$$

which is to say that:

$$|AA_b| = |C_aC| + |C_bC|$$

we have for the lengths of the radii $r_{a,b,c}$ of the circles $s_{a,b,c}$:

$$2\sqrt{r_b \cdot r_a} = 2\sqrt{r_a \cdot r_c} + 2\sqrt{r_b \cdot r_c}$$

Dividing the above relation through by $2\sqrt{r_a \cdot r_b \cdot r_c}$, we arrive at a relationship that ties the lengths sought-after together:

$$\boxed{\frac{1}{\sqrt{r_c}} = \frac{1}{\sqrt{r_a}} + \frac{1}{\sqrt{r_b}}} \tag{1}$$

which is quite neat and is quite recursive because into the curvilinear triangle trapped between the circles $s_{a,c}$ or the circles $s_{b,c}$ and the straight line l, using the methodology already explained, yet another circle, s_d, such that:

$$\frac{1}{\sqrt{r_d}} = \frac{1}{\sqrt{r_c}} + \frac{1}{\sqrt{r_b}} = \frac{1}{\sqrt{r_a}} + \frac{2}{\sqrt{r_b}}$$

can be inscribed.

By playing with the carefully crafted numbers, the numbers whose square roots can be perfectly extracted, it is possible to arrive at some interesting sequences (or series) of numbers which, seemingly, have nothing to do with Euclidean geometry whatsoever. For example, by setting $r_b = 1$ and $r_a = 1/4$, we see that:

$$r_c = \frac{1}{9} = \frac{1}{3^2}$$

and that:

$$r_d = \frac{1}{16} = \frac{1}{4^2}$$

If we inscribe another circle, s_e, that touches the circles $s_{d,b}$ then:

$$r_e = \frac{1}{25} = \frac{1}{5^2}$$

and so on. With the chosen values of $r_{a,b}$, the lengths of the radii of such circles will probe the reciprocal squares of consecutive whole positive numbers. Summed indefinitely, numerically the lengths of these radii will match the value of the Riemann Zeta function evaluated at the argument 2:

$$\zeta(2) = \sum_{n=1}^{+\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

If we were to consecutively inscribe the circles in such a way that they would always touch the circle s_a instead then the lengths of the radii of such circles will probe the reciprocal squares of consecutive **odd** whole positive numbers:

$$\sum_{n=0}^{+\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

and so on.

We may also inflate the problem by one more dimension, in which case the tangent circles $s_{a,b,c}$ can actually be the tangent *spheres* and the straight line l can actually be a *plane*. $\square$